

Lecture 6: Workspace Setup Checklist

Follow this checklist to verify you completed each step of the walkthrough.

Part I: Software Installation

- ☐ R version 4.0+ confirmed (`R.version.string`)
- ☐ `remotes` package installed
- ☐ `devtools` package installed
- ☐ `usethis` package installed
- ☐ `renv` package installed
- ☐ `testthat` package installed
- ☐ `palmerpenguins` package installed
- ☐ `tidyverse` package installed
- ☐ `rrtools` package installed from GitHub
- ☐ All packages load without errors

Part II: Create Research Compendium

- ☐ Working directory set to a stable location (not Dropbox/OneDrive)
- ☐ Compendium created with `rrtools::use_compendium("penguins.yourname")`
- ☐ New RStudio window opened automatically
- ☐ Analysis structure added with `rrtools::use_analysis()`
- ☐ Folder structure verified with `fs::dir_tree()`

Expected structure:

```
penguins.yourname/
├── DESCRIPTION
├── analysis/
│   ├── data/
│   │   ├── raw_data/
│   │   └── derived_data/
│   ├── figures/
│   └── paper/
│       └── paper.Rmd
└── R/
```

Part III: Add Data

- ☐ Palmer penguins data exported to CSV:
`write.csv(penguins, "analysis/data/raw_data/penguins.csv", row.names = FALSE)`
- ☐ Data verified: 344 rows, 8 columns

Part IV: Add Analysis Plan Template

- ☐ `analysis_plan_template.qmd` copied from Lecture 5
- ☐ File saved to `analysis/paper/analysis_plan.qmd`
- ☐ File exists: `file.exists("analysis/paper/analysis_plan.qmd")` returns TRUE

Part V: Write Unit Tests

- ☐ `testthat` initialized with `usethis::use_testthat()`
- ☐ `tests/testthat/` folder exists

Test 1: Data Validation

- ☐ Test file created: `usethis::use_test("data-validation")`
- ☐ Test code added to `tests/testthat/test-data-validation.R`:

```
test_that("penguins data file exists and has correct dimensions", {  
  data_path <- "analysis/data/raw_data/penguins.csv"  
  expect_true(file.exists(data_path))  
  penguins_data <- read.csv(data_path)  
  expect_equal(nrow(penguins_data), 344)  
  expect_equal(ncol(penguins_data), 8)  
})
```

Helper Function

- ☐ Function file created: `usethis::use_r("clean-data")`
- ☐ Function added to `R/clean-data.R`:

```
#' Remove rows with missing values from penguins data  
#'  
#' @param data A data frame of penguin measurements  
#' @return A data frame with complete cases only  
clean_penguins <- function(data) {  
  data[complete.cases(data), ]  
}
```

Test 2: Function Validation

- ☐ Test file created: `usethis::use_test("clean-data")`
- ☐ Test code added to `tests/testthat/test-clean-data.R`:

```
test_that("clean_penguins removes rows with missing values", {  
  test_data <- data.frame(  
    species = c("Adelie", "Gentoo", "Chinstrap"),  
    bill_length = c(39.1, NA, 46.5),  
    body_mass = c(3750, 4500, NA)  
  )  
  cleaned <- clean_penguins(test_data)  
  expect_equal(nrow(cleaned), 1)  
  expect_equal(cleaned$species, "Adelie")  
  expect_false(any(is.na(cleaned)))  
})
```

Run Tests

- ☐ Functions loaded: `devtools::load_all()`
- ☐ Tests run: `devtools::test()`
- ☐ Both tests pass (green checkmarks)

Part VI: Version Control and GitHub

- ☐ Git initialized: `usethis::use_git()`
- ☐ Initial commit made
- ☐ renv initialized: `renv::init()`
- ☐ Packages snapshot taken: `renv::snapshot()`
- ☐ `renv.lock` file exists
- ☐ renv files committed (`renv.lock`, `renv/`, `.Rprofile`)

- ☐ GitHub repository created: `usethis::use_github(private = FALSE)`
- ☐ Browser opened to your new repository

Part VII: Final Verification

- ☐ Project structure verified with `fs::dir_tree(recurse = 2)`
- ☐ `devtools::check()` runs with 0 errors

Final Project Structure

Your project should look like this:

```
penguins.yourname/
├── DESCRIPTION
├── NAMESPACE
├── .Rprofile
├── renv.lock
├── analysis/
│   ├── data/
│   │   ├── raw_data/
│   │   │   └── penguins.csv
│   │   └── derived_data/
│   ├── figures/
│   └── paper/
│       ├── analysis_plan.qmd
│       └── paper.Rmd
├── R/
│   └── clean-data.R
├── renv/
│   └── ...
└── tests/
    ├── testthat.R
    └── testthat/
        ├── test-clean-data.R
        └── test-data-validation.R
```

Next Steps

Before January 27:

1. **Fill in your analysis plan** - Edit `analysis/paper/analysis_plan.qmd`
2. **Commit changes regularly** - Use Git pane in RStudio
3. **Push to GitHub** - Keep your repository updated

Troubleshooting

“rrtools not found”

```
remotes::install_github("benmarwick/rrtools")
```

“GitHub PAT not found”

```
usethis::create_github_token()
gitcreds::gitcreds_set()
```

“Test failed”

1. Check file paths are correct
2. Make sure you saved all files
3. Run `devtools::load_all()` before `devtools::test()`

“could not find function ‘clean_penguins’”

```
devtools::load_all()
```

“renv::restore() fails”

```
renv::clean()  
renv::restore()
```

Questions?

- **Office hours:** By appointment
- **Email:** rgthomas@ucsd.edu or maluo@ucsd.edu
- **Canvas:** Post in discussion board